ANOVA decomposition and hypothesis testing:

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LSM

Recall $\mathcal{M}_1 : Y = Z\gamma + W\delta + \varepsilon$ ⟵ full model

$\mathcal{M}_2 : Y = Z\gamma + \varepsilon$

$\mathcal{M}_2$ corresponds to $H_0 : \delta = 0$

Rejecting $H_0 : \delta = 0$ in favour of $H_1 : \delta \neq 0$ is to say that $\mathcal{M}_2$ is not adequate (in favour of $\mathcal{M}_1$)

Note: Full model may not be true either
- linearity may not hold
- some relevant covariate may be missing

A justification for G-M model with Gaussian noise comes from the assumption that (X_i, Y_i) are i.i.d. sample from a multivariate normal distribution

LRT for $H_0 : \delta = 0$ vs. $H_1 : \delta \neq 0$ is equivalent to rejecting H_0 for large values of $F = \dfrac{\frac{SSE_{red} - SSE_{full}}{df(\;''\;)}}{\frac{SSE_{full}}{df(\;''\;)}}$

Associated ANOVA Decomposition

$$Y^T Y = Y^T P_X Y + Y^T (I - P_X) Y$$

$$= Y^T (P_Z + P_{W|Z}) Y + \underbrace{Y^T (I - P_X) Y}_{SSE_{full}}$$

$$SSE_{red} = Y^T (I - P_Z) Y$$

$$SSE_{red} - SSE_{full} = Y^T P_{W|Z} Y$$

$$df(SSE_{red} - SSE_{full}) = \operatorname{rank}(P_{W|Z})$$

$$df(SSE_{full}) = \operatorname{rank}(I - P_X) = n - \operatorname{rank} P_X$$

$$\text{rank}(X) = \text{rank}(Z) + \text{rank}((I-P_Z)W)$$

Special case: Variable selection

$$Y_i = \beta_0 + \beta_1 X_i^{(1)} + \dots + \beta_{p-1} X_i^{(p-1)} + \varepsilon_i$$

Want to know whether variables $X^{(q)}, \dots, X^{(p-1)}$ are important
$q \le p-1$

$$H_0: \beta_q = \dots = \beta_{p-1} = 0$$

$$R^2 = \frac{SSR_{full}}{SST_0} = \frac{Y^T(P_X - P_{\underset{\sim}{1}})Y}{Y^T(I - P_{\underset{\sim}{1}})Y} \qquad X = [\underset{\sim}{1}\ \underset{\sim}{X}^{(1)} \dots \underset{\sim}{X}^{(p-1)}]$$

↑
multiple R^2

What is partial correlation coefficient

Partial Correlation b/w Y and $X^{(j)}$
is the correlation b/w residuals $(I-P_{X_{-j}})Y$
of fitting linear regression of Y and $\{X^{(k)}\}_{k\neq j}$
and residuals of fitting linear regression of
$X^{(j)}$ on $\{X^{(k)}\}_{k\neq j} \longrightarrow (I-P_{X_{-j}})X^{(j)}$

Let $\underset{n\times(p-1)}{X_{-j}} = [\underset{\sim}{1}\ \underset{\sim}{X}^{(1)} \dots \underset{\sim}{X}^{(j-1)}\ \underset{\sim}{X}^{(j+1)} \dots \underset{\sim}{X}^{(p-1)}]$

So partial covariance $= \frac{1}{n}((I-P_{X_{-j}})Y)^T(I-P_{X_{-j}})\underset{\sim}{X}^{(j)}$

$$= \frac{1}{n} Y^T(I-P_{X_{-j}})\underset{\sim}{X}^{(j)}$$

$\Rightarrow$ partial correlation coefficient $= \dfrac{Y^T(I-P_{X_{-j}})\underset{\sim}{X}^{(j)}}{\sqrt{Y^T(I-P_{X_{-j}})Y}\sqrt{\underset{\sim}{X}^{(j)T}(I-P_{X_{-j}})\underset{\sim}{X}^{(j)}}}$

Squared partial correlation coefficient

$$\frac{\left(Y^T(I-P_{X_{-j}})X^{(j)}\right)^2}{\left(Y^T(I-P_{X_{-j}})Y\right)\left(X^{(j)T}(I-P_{X_{-j}})X^{(j)}\right)}$$

$\hat{\beta}_j$ in the full model

$$\hat{\beta}_j = \frac{Y^T(I-P_{X_{-j}})X^{(j)}}{X^{(j)T}(I-P_{X_{-j}})X^{(j)}} \sim N\left(\beta_j, \frac{\sigma^2}{X^{(j)T}(I-P_{X_{-j}})X^{(j)}}\right)$$

$$P_X - P_{X_{-j}} = P_{(I-P_{X_{-j}})X^{(j)}}$$

Result: Let $\underset{n\times p}{X} = [\underset{n\times r}{Z} : \underset{n\times q}{W}]$ $\quad p = q + r$, $q = \text{rank}(W)$, $r = \text{rank}(Z)$

Define $U = \dfrac{Y^T(I-P_X)Y}{Y^T(I-P_Z)Y} = \dfrac{SSE_{full}}{SSE_{red}}$ (reduced $\leftrightarrow \delta = 0$)

Result

ind $\begin{cases} V_1 \sim \text{Gamma}(a) \\ V_2 \sim \text{Gamma}(b) \end{cases}$

$$\frac{V_1}{V_1+V_2} \sim \text{Beta}(a,b)$$

$$= \frac{SSE_{full}}{SSE_{full} + (SSE_{red} - SSE_{full})} \sim \text{Beta}\left(\frac{n-p}{2}, \frac{p-r}{2}\right)$$

$$Y^TY = n\bar{Y}^2 + \sum(Y_i - \bar{Y})^2 = Y^T P_{\underset{\sim}{1}} Y + Y^T(I - P_{\underset{\sim}{1}})Y$$

Fisher-Cochran Theorem:

Let $\begin{cases} Y \sim N(\mu, I_n) \\ A_1, \dots, A_r \text{ be symm. positive semidefinite} \end{cases}$

$$Y^TY = \sum_{j=1}^{r} Y^T A_j Y \quad \text{for all } y \in \mathbb{R}^n$$

Then ① $Y^T A_j Y \overset{\text{indep}}{\sim} \chi^2_{\rho(A_j)}\left(\frac{\lambda_j}{2}\right)$ for $j=1,\dots,r$

if and only if ② $\sum_{j=1}^{r} \rho(A_j) = n$

in which case $\lambda_j = \mu^T A_j \mu$ for $j=1,\dots,r$

and $\mu^T\mu = \sum \lambda_j$

① $\Rightarrow$ ② $\exists X_1, \dots, X_r$ with $X_j \sim N(\delta_j, I_{\rho(A_j)})$

s.t. $X_j^T X_j \sim \chi^2_{\rho(A_j)}\left(\frac{\|\delta_j\|^2}{2}\right)$

Then $X_j^T X_j \overset{D}{=} Y^T A_j Y$

$X = \begin{pmatrix} X_1 \\ \vdots \\ X_r \end{pmatrix}$ and $X^TX = \sum_{j=1}^{r} X_j^T X_j = \sum_{j=1}^{r} Y^T A_j Y = Y^TY$

Thus on one hand

$$Y^TY \overset{D}{=} X^TX \sim \chi^2_{\sum_{j=1}^{r}\rho(A_j)}\left(\frac{\sum_{j=1}^{r}\lambda_j}{2}\right)$$

$$Y^TY \sim \chi^2_n\left(\frac{\|\mu\|^2}{2}\right)$$

$\therefore \sum_{j=1}^{r} \rho(A_j) = n, \quad \sum_{j=1}^{r} \lambda_j = \mu^T\mu$

also $\mathbb{E}(Y^T A_j Y) = \mathbb{E}\, tr(Y^T A_j Y)$

$$= \mathbb{E}\, tr(A_j Y Y^T)$$

$$= tr\, \mathbb{E}(A_j Y Y^T)$$

$$= tr(A_j \mathbb{E}(Y Y^T))$$

$$= tr(A_j(\mu\mu^T + I))$$

$$= tr(A_j) + \mu^T A_j \mu$$

$$\lambda_j = \mathbb{E}(Y^T A_j Y) - \rho(A_j) \overset{?}{=} \mu^T A_j \mu$$

We note that $\mathbb{E}(Y^T A_j Y) = \mu^T A_j \mu + tr A_j$

and $\lambda_j = \mathbb{E}(Y^T A_j Y) - \rho(A_j)$

$$= \mu^T A_j \mu + \underbrace{tr A_j - \rho(A_j)}_{=0} \quad \text{—— } (*)$$

Proof 2 $\Rightarrow$ 1

Since for each j, each A_j is positive semidefinite there exist $B_1, \dots, B_r$ s.t. $B_{j\, n\times\rho(A_j)}$ with $\rho(B_j) = \rho(A_j)$ s.t. $A_j = B_j B_j^T$ and so

$$\sum_{j=1}^{r} A_j = \sum_{j=1}^{r} B_j B_j^T = BB^T \text{ where } B_{n\times n} = [B_1 : \dots : B_r]$$

$$\left[\because \sum_{j=1}^{r} \rho(A_j) = n\right]$$

Want to show that $BB^T = I$

We know $y^T BB^T y = y^T y$

$\Rightarrow$ all the eigenvalues of the symmetric matrix BB^T are 1

$\Rightarrow BB^T = I_n$

$$\Rightarrow B^T B = I_n = \begin{bmatrix} B_1^T \\ B_2^T \\ \vdots \\ B_r^T \end{bmatrix} \begin{bmatrix} B_1 & B_2 & \cdots & B_r \end{bmatrix}$$

$$\Rightarrow B_j^T B_j = I_{\rho(A_j)}$$

$\Rightarrow A_j = B_j B_j^T$ is an orthogonal projection matrix

$$\left(B_j \underline{B_j^T B_j} B_j^T = B_j B_j^T \right)$$

$\downarrow$

Note: Hence Ⓐ is true

<u>Exercise</u>: Suppose $Y \sim \mathcal{N}(X\beta, \sigma^2 I_n)$

Let $L_{p \times k}$ with $\mathrm{rank}(L) = k$, $\mathrm{rank}(X) = r$

and $\mathcal{C}(L) \subset \mathcal{C}(X^T) = \mathcal{R}(X)$

Define
$$S_0 = Y^T (I - P_X) Y$$
$$S_1 = \min_{\beta : L^T\beta = \theta_0} \|Y - X\beta\|^2$$

Then
$$\frac{S_0}{\sigma^2} \sim \chi^2_{n-r}$$
$$\frac{S_1 - S_0}{\sigma^2} \sim \chi^2_k(\delta)$$ with some non-centrality parameter

$\delta = 0$ if $L^T\beta = \theta_0$